

Cleaning Report



OF Adult Census Income Analysis

Introduction

Data cleaning is an important step in data analysis. It improves the quality of the dataset by identifying and correcting missing values, duplicate records, and other inconsistencies. A clean dataset helps in producing accurate analysis and reliable results.

Data Cleaning Steps

1. Loaded the Dataset

The Adult Census Income dataset was loaded into a Pandas DataFrame using the `read_csv()` function.

2. Examined the Dataset

The dataset was examined using the following functions:

- `head()` – Displayed the first five rows.
- `tail()` – Displayed the last five rows.
- `info()` – Displayed information about the dataset, including data types and non-null values.
- `shape` – Displayed the number of rows and columns.
- `columns` – Displayed the names of all columns.

This helped in understanding the structure of the dataset before cleaning.

3. Checked Missing Values

Missing values were identified using the `isnull().sum()` function.

Since the dataset stored missing values as "?", they were first replaced with NaN using the `replace()` function so that Pandas could recognize them as missing values.

4. Filled Missing Values

The missing values in the following categorical columns were filled using the **mode (most frequent value)** of each column:

- **workclass** → Filled with **Private**
- **occupation** → Filled with **Prof-specialty**
- **native.country** → Filled with **United-States**

Using the mode ensured that no records were removed and the dataset remained complete.

5. Checked Duplicate Records

Duplicate rows were identified using the `duplicated().sum()` function. A total of **24 duplicate rows** were found in the dataset.

6. Removed Duplicate Records

The duplicate rows were removed using the `drop_duplicates()` function. After removal, the dataset contained **32,537 rows** and **15 columns**.

7. Verified Data Types

The data types of all columns were checked using the `dtypes` function to ensure that numerical columns and categorical columns were stored in the correct format.

8. Generated Statistical Summary

The `describe()` function was used to generate summary statistics for all numerical columns. It provided information such as count, mean, standard deviation, minimum value, maximum value, and quartiles, helping to understand the distribution of the numerical data.

Summary

➔ After completing the data cleaning process:

- Missing values were successfully identified and filled using the mode.
- Duplicate records were removed.
- Data types were verified.
- The dataset was checked using descriptive statistics.
- The final cleaned dataset contained **32,537 rows** and **15 columns** and was ready for Exploratory Data Analysis (EDA).